A

1. operators are used to evaluate expressions, compare conditions and all

   Types: a. Arithmetic b. Logical c. Assignment d. Comparison

2. `= is used for assignment`

   `== is used for comparison`

3. input() function is used to take input from user at terminal

   ex:

   name = input("Enter your name : ")

4. for loop is used to repeated task when we know the exact iterations

while loop is used when we don't know the iteration but we know the condition to stop

5. string slicing is used to exact a specific portion from a string

   syntax : new = old[start: end: step]

   ex : name = "Rutik Bajad" fname = name[:5]

6. len() is used to get the lenght of string or collections.

   name = "Rutik"

   print(len(name))

7. when a string is reversed and it is same as the original one then this condition is called palindrome

   ex : madam

8. pass is a keyword that is used when we write functions, loops, and conditions but we don't want to write the code for right now and don't want the error to

   so pass will escape error

9. easy to understand and write

   large community support

   faster for development

B

10.

4

1

11.

Pyth

hon

12.

2 4 6 8

13.

5

4

3

2

1

14.

rotneMneveS

evenMen

C

In [2]:
```
# 15.
name = "Rutik"
print(len(name))
```

```
5
```

In [4]:
```
# 16.
name = "Rutik"
rev = ""
for ch in name:
    rev = ch + rev
rev
```

Out[4]: 'kituR'

In [5]:
```
# 17.
str1 = "Rutik"
str2 = "Rutik"
equal = True
for i in range(len(str1)):
    if str1[i] != str2[i]:
        equal = False
        break
if equal:
    print("Both strings are equal")
```

```
Both strings are equal
```

In [6]:
```
# 18.
name = "Rutik"
for ch in name:
    print(ch)
```

```
R
u
t
i
k
```

D

In [8]:
```
# 19.
n = int(input())
if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

```
Positive
```

In [9]:
```
# 20.
age = int(input("Enter your age: "))
if age >= 18:
    print("You can vote.")
else:
    print("You cannot vote.")
```

```
You cannot vote.
```

In [11]:
```
# 21.
a, b, c = map(int, input().split())
if a >= b and a >= c:
    print(f"{a} is largest.")
elif b >= a and b >= c:
    print(f"{b} is largest.")
else:
    print(f"{c} is largest")
```

```
6 is largest
```

E

In [12]:
```
# 22.
n = int(input())
for i in range(1, n+1):
    print(i, end="\t")
```

```
1	2	3	4	5	6	7
```

In [13]:
```
# 23.
n = int(input())
i = 2
while i != n:
    if i % 2 == 0:
        print(i, end="\t")
    i += 1
```

```
2	4	6	8	10	12	14	16	18	20	22	24
```

F

In [14]:
```
# 24.
n = int(input())
for i in range(1, n+1):
    print("*"*i)
```

```
*
**
***
****
*****
```

In [16]:
```
# 25.
n = int(input())
for i in range(1, n+1):
    print(f"{" "*(n-i)}{"*"*i}")
```

```
    *
   **
  ***
 ****
*****
```

In [5]:
```
# 26.
n = int(input())
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end="")
    num += 1
    print()
```

```
1
22
333
4444
55555
```

In [14]:
```
# 27.
n = int(input())
for i in range(n, 0, -1):
    for j in range(i):
        print(n, end="")
    n -= 1
    print()
```

```
55555
4444
333
22
1
```

Challenge Questions

In [16]:
```
# 28.
n = int(input())
a, b, c = 0, 1, 1
for i in range(n):
    print(a, end="\t")
    a, b, c = b, c, a+b+c
```

```
0	1	1	2	4	7	13	24	44	81
```

In [21]:
```
# 29.
n = int(input())
k = 1
for i in range(n+1):
    for j in range(i):
        print(k, end=" ")
        k += 1
    print()
```

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

In [30]:
```
# 30.
for i in range(1, 5):
    for j in range(1, 7):
        if i == 1 or j == 1 or i == 4 or j == 6:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

```
******
*    *
*    *
******
```

Bonus Questions

31.

for loop is used when we know the no. of iterations

while loop is used when we know the condition to stop

In [24]:
```
# 32.
i = 1
while i <= 10:
    print(i, end="\t")
    i += 1
```

```
1	2	3	4	5	6	7	8	9	10
```